External Validation of a Seven-Weight Scorecard versus Tree Ensembles in Ovarian Cancer Risk Prediction

Jiayi Yao
BASIS Independent Bellevue
Bellevue, WA, USA
High School Student
yaojiayi2020@gmail.com

Zooey Lane Go Hua
Issaquah High School
Issaquah, WA, USA
High School Student
Zooey.hua@gmail.com

Vritika S Sharma
Issaquah High School
Issaquah, WA, USA
High School Student
vritikas01@gmail.com

***Abstract*—Preoperative blood tests help decide whether an ovarian mass requires cancer surgery; a model that degrades silently at a new hospital can delay surgery or trigger unnecessary operations. This paper asks whether more complex machine learning models transfer better to new hospitals than simple ones. Eight models (a CA125 cutoff rule, the clinical formulas ROMA and CPH-I, a logistic regression, an integer scorecard, and three tree ensembles) were trained on a Chinese dataset (349 patients) and tested, unchanged, on two other Asian cohorts (West China, 380 patients; Japan, 177). On the 100 West China patients with measured biomarkers, the scorecard (AUROC 0.971) was not significantly different from the logistic regression (0.941) and ROMA (0.965), and was numerically the highest. A simulation calibrated to the real data, intended only as an illustrative counterexample, showed that under the simulated concept drift examined here the flexible models' advantage diminished and reversed. Within these small Asian cohorts, the scorecard performed no worse than far more complex ensembles and, at its fixed high-risk tier, missed fewer cancers than the published formulas at their cutpoints; generalization beyond these cohorts requires larger validation studies.**

***Keywords*—ovarian cancer; risk prediction; scorecard; external validation; transportability**

I. Introduction

Ovarian cancer is the most lethal gynecologic cancer, and most cases are found late. When a pelvic mass is found, whether to send the patient to a specialist cancer surgeon depends on preoperative risk, and getting it wrong in either direction is costly. That decision currently relies on two blood proteins, CA125 (cancer antigen 125) and HE4 (human epididymis protein 4), measured cheaply before surgery and combined either by fixed formulas such as the Risk of Ovarian Malignancy Algorithm (ROMA) [1] and the Copenhagen Index (CPH-I) [2], or by a fitted model. Machine learning models have been proposed for the same task, usually with strong results on the data they were trained on but little testing at other hospitals [3]–[5].

Lu et al. [6] trained machine-learning models on the Changzhou cohort reused here, selecting HE4 and CEA for a decision tree that beat ROMA and logistic regression on a held-out split. Our work adds a hand-computable scorecard, applies every model frozen to two Asian cohorts, and asks how much of such an advantage survives the move to a new hospital.

This paper asks two linked questions. First, does model complexity translate into performance that survives a move to a new hospital? Second, can a model simple enough to be computed at the bedside (an integer scorecard) match the more complex alternatives? Interpretability arguments favor such simple models in high-stakes medicine [7], [8]. We trained eight models on one Chinese cohort, applied them unchanged to two independent Asian cohorts, and measured discrimination, calibration, and decision consequences, plus a simulation of when complex models help.

II. Methods

A. Data

Three public, anonymized datasets were used [9]–[11]. Institutional review board approval was not required for re-analysis of public, de-identified data; the cohorts are a training cohort from Changzhou, Jiangsu, China (349 patients; 171 ovarian cancers, 178 benign masses; Mendeley Data [9], doi:10.17632/th7fztbrv9.11, from the study by Lu et al. [6]), a second Chinese cohort from the West China Second University Hospital of Sichuan University, Chengdu (380 patients; 188 cancers, 192 benign cysts; figshare [10], doi:10.6084/m9.figshare.28831256), and a Japanese cohort (177 patients; 27 cancers, 150 healthy women; Karger figshare [11], doi: 10.6084/m9.figshare.24235450). Each patient has four features: CA125 (U/mL), HE4 (pmol/L), age, and menopausal status. The source files are `chinese training dataset(All Raw Data).csv`, `west_china.xlsx`, and `japanese.xlsx`; `src/clean_data.py` harmonizes them into the three processed files used by every analysis. In the training cohort, scattered missing biomarker values were filled by each feature's cohort median. Externally, HE4 was not measured for 74% of West China patients and age was not recorded for 85% of Japan patients; these external gaps were filled with the frozen training-set medians, so no statistic is computed on a validation cohort. Because the Japanese cohort contains 150 healthy women and only 27 cancers, and algorithm performance depends on the malignancy base rate among the patients tested [12], it is treated as a specificity stress test, not external validation: its AUROC is reported separately and never pooled with West China's. The primary West China analyses use the 100 patients with measured HE4 (63 cancers, 37 benign), so that the external result rests on measured biology rather than imputed constants; the 280 excluded patients (125 cancers, 155 benign) had similar age, CA125, and menopausal rates.

B. Sensitivity to Missing Biomarkers

The primary external benchmark uses the 100 West China patients with measured HE4. The 74%-imputed full cohort ($N = 380$) is reported as a sensitivity analysis, because median imputation may alter model discrimination; no-HE4 refits and MICE AUROCs bound this effect (Section III-C).

C. Models

Eight models used the same four features. (1) *Integer scorecard*: each continuous feature was rank-transformed and cut into training-set tertiles (frozen cutpoints: CA125 24.3/165.0 U/mL, HE4 45.8/92.7 pmol/L, age 37.0/52.0 years), then one-hot encoded and fitted with an L2-regularized logistic regression ($C = 1.5$); each coefficient β_i was then rescaled to the nearest whole number $w_i = \text{round}(5 \times \beta_i / \max_j |\beta_j|) \in \{-5, \ldots, +5\}$, with $|\beta_i| < 0.01$ set to zero. The score is the sum of these feature points (range -7 to $+9$), can be computed by hand (Table I), and contains seven fitted integer parameters (two free bin weights per continuous feature plus an unused intercept). Its weights match pathophysiology: HE4 spans -4 to $+5$ versus -1 to $+2$ for CA125, matching HE4's higher cancer specificity and CA125's elevation in endometriosis; older age adds points. The regularized fit assigned the binary menopausal feature a coefficient that rounded to zero, so it is effectively dropped (its row in Table I documents this). (2) *Logistic regression* on standardized features. (3–4) the published formulas *ROMA* and *CPH-I*, computed on raw values as follows. ROMA uses PI = $-12.0 + 2.38 \ln(\text{HE4}) + 0.0626 \ln(\text{CA125})$ premenopause and PI = $-8.09 + 1.04 \ln(\text{HE4}) + 0.732 \ln(\text{CA125})$ postmenopause, mapped to a percentage by the logistic transform, with published cutoffs 13.1% (pre) and 27.7% (post) [1]. CPH-I uses $-14.0647 + 1.0649 \log_2(\text{HE4}) + 0.6050 \log_2(\text{CA125}) + 0.2672 \times \text{age}/10$, likewise logistic-transformed, with cutoff 7% [2]. Units: HE4 pmol/L, CA125 U/mL, age years; menopause is the dataset's binary postmenopausal indicator; imputed HE4 uses the frozen training median. Within-cohort recalibration means fitting a logistic regression of the outcome on the formula's probability as the sole covariate, on West China. (5) a single *CA125 ≥ 35 U/mL* rule. (6–8) *XGBoost* [13], *CatBoost* [14], and *Random Forest* [15], evaluated under two settings: Setting A keeps each package's default hyperparameters, and Setting B tunes each ensemble by nested cross-validation, with all tuning decisions confined to training folds (XGBoost 100 trees, depth 3; CatBoost 500 iterations, depth 4; Random Forest 500 trees, depth 6). The primary scientific comparison uses Setting B (Table II, Fig. 1); Setting A values are in Table II's caption. All models were trained once on the Chinese cohort, and all parameters, bin cutpoints, and thresholds were frozen before any external evaluation; the models were applied unchanged everywhere else.

TABLE I
THE INTEGER SCORECARD, FITTED ON THE CHINESE TRAINING COHORT (349 PATIENTS), APPLIED UNCHANGED EXTERNALLY; FEATURES BINNED BY FROZEN TRAINING-SET TERTILES; THE BINARY MENOPAUSAL FEATURE RECEIVED A COEFFICIENT THAT ROUNDED TO ZERO.

Feature	Low bin	Mid bin	High bin
CA125 (U/mL)	−1	0	+2
HE4 (pmol/L)	−4	−1	+5
Age (years)	−2	0	+2
Menopausal status	0	0	0

D. Evaluation

Discrimination was measured by the area under the ROC curve (AUROC), with 95% confidence intervals and two-sided paired bootstrap tests between models from 2,000 patient-level resamples. The prespecified primary comparison is the scorecard versus ROMA on West China; all other comparisons are secondary, and effect sizes with 95% CIs are emphasized over p-values, with Holm–Bonferroni adjustment across the six comparisons of Section III-B. Calibration was measured with the Brier score and calibration-in-the-large [16]. Decision value was quantified with a weighted-harm analysis (Section III-B) [17]: thresholds were swept over 1–100%, and for exchange rates $w \in \{1, 5, 10, 50\}$ the threshold minimizing $H = w \times \text{missed} + \text{unnecessary}$ was identified. We distinguish Q1 (does each model transport unchanged?) from Q2 (how well can each model perform after local recalibration?). The exchange-rate weights are hypothetical scenarios, not validated clinical utilities: they span from valuing a missed cancer and an unnecessary referral equally ($w = 1$) to valuing a missed cancer 50 times as much ($w = 50$), characterizing sensitivity to relative error costs rather than establishing clinical utility. All design choices preceded external evaluation: the features are those common to the cohorts and required by the published formulas; the primary comparison and the four weights were pre-specified; only the scorecard's bin count and regularization were selected among 12 candidates by repeated CV (Section III-E). The primary comparison applies the same within-cohort recalibration to every model; published-cutpoint operating points are reported separately; no pooled decision analysis was performed. The retraining experiment used stratified training subsets ($n = 60$ to 349, 10 draws each), evaluated unchanged on the imputed West China cohort. Robustness was tested with 10% and 30% multiplicative, additive, and combined noise, 5% gross-error contamination, and ±10% bin-boundary shifts.

E. Implementation and availability

Analyses used Python 3.13 with scikit-learn 1.7.2, XGBoost 3.3.0, CatBoost 1.2.10, and Random Forest, all with random seed 42; cross-validation used stratified five-fold splits, and the simulation used 20 replicates per condition. Setting A (package defaults): XGBoost 100 trees, depth 6, learning rate 0.3; CatBoost 1,000 iterations, depth 6, learning rate 0.03; Random

Forest 100 trees, no depth cap. Setting B (primary): the nested-CV-selected configurations of Section II-C. All preprocessing, model fitting, evaluation, figures, and sensitivity analyses are implemented in a reproducible pipeline (`reproduce.sh`) with pinned dependencies and a fixed random seed, available at https://github.com/angelayaoo/Ovarian-Cancer-RiskPrediction, following calls for code release [18].

F. Simulation

The simulation is an illustrative counterexample, not an explanation of the observed real-world results: it varies drift (d) independently of covariate shift (s) and noise. A simulation calibrated to the Chinese data drew biomarkers from class-conditional lognormal distributions and generated the cancer label from the logistic model

$$\begin{aligned}\eta(z_1, z_2, z_3) = &-0.65 + 1.8z_1 + 1.0z_2 + 0.5z_3 \\ &+ (1-d)\big(1.6z_1z_2 + 1.2\sin(1.7z_1)\big),\end{aligned} \tag{1}$$

where z_1, z_2, and z_3 are the standardized log CA125, log HE4, and age, respectively, and $d \in [0, 1]$ sets the drift level: $d = 0$ keeps the interaction and nonlinear terms, $d = 1$ removes them from the test-side model, and intermediate values scale them by $1 - d$, as if the biomarker–cancer relationship partially changed between hospitals. The grid crossed training sizes $n \in \{100, 200, 400, 800\}$, distribution shifts $s \in \{0, 0.5, 1.0\}$, measurement noise $\{0\%, 50\%\}$, and drift levels $d \in \{0, 1\}$, with 20 replicates per cell.

III. Results

A. Discrimination

On the 100 West China patients with measured HE4 (63 cancers), the scorecard was numerically highest, with ROMA, tuned XGBoost, and the logistic regression within overlapping CIs; the rest were lower (Table II, West CC column; Fig. 1).

On the 74%-imputed full cohort ($N = 380$; sensitivity analysis), the ranking was similar (logistic regression 0.936, scorecard 0.929, tuned ensembles 0.877–0.895; Table II, West column).

On the imputed West China cohort, the scorecard-minus-ROMA difference was +0.043 AUROC (95% CI 0.012 to 0.073; $p = 0.008$); the scorecard also exceeded CPH-I (+0.061; 0.032 to 0.092), XGBoost (+0.040; 0.012 to 0.068), CatBoost (+0.052; 0.021 to 0.086), and Random Forest (+0.034; 0.006 to 0.065), and did not differ from the logistic regression (−0.007; −0.025 to 0.011; $p = 0.48$). After Holm–Bonferroni correction across the six comparisons, the scorecard remained significantly ahead of CPH-I, XGBoost, and CatBoost (adjusted $p < 0.01$), ROMA (0.02), and Random Forest (0.03).

In the Japan specificity stress test ($n = 177$), CPH-I was highest (0.920; Table II, Japan column). Within Japan, 86% of healthy women but only 30% of cancer patients fell into both markers' lowest bins; West China had higher CA125 (65.1 vs. 49.3 U/mL), the volunteers far lower markers (CA125 15.1, HE4 33.8).

TABLE II

AUROC (95% CI) by model and cohort, bootstrap means over 2,000 resamples. West CC: the 100 West China patients with measured HE4 (primary external benchmark); West: the full 74%-imputed cohort ($N = 380$, sensitivity analysis); Japan: healthy volunteers (specificity stress test). Setting A (defaults) ensembles scored 0.921/0.922/0.910 (CC), 0.887/0.879/0.879 (West), 0.734/0.749/0.792 (Japan). Bold: highest per cohort, or indistinguishable from it.

Model	West CC (n=100)	West (N=380)	Japan
Scorecard	**0.971** **(0.936–0.995)**	**0.929** **(0.902–0.955)**	0.845 (0.714–0.949)
Logistic reg.	0.941 (0.887–0.983)	**0.936** **(0.908–0.961)**	0.806 (0.666–0.926)
ROMA	0.965 (0.930–0.992)	0.886 (0.848–0.921)	0.799 (0.666–0.908)
CPH-I	0.929 (0.875–0.973)	0.868 (0.828–0.905)	**0.920** **(0.852–0.973)**
CA125 rule	0.694 (0.605–0.785)	0.642 (0.597–0.684)	0.781 (0.685–0.874)
XGBoost	0.946 (0.897–0.984)	0.890 (0.853–0.925)	0.840 (0.711–0.944)
CatBoost	0.910 (0.845–0.962)	0.877 (0.836–0.915)	0.772 (0.650–0.880)
Random Forest	0.930 (0.871–0.979)	0.895 (0.858–0.931)	0.819 (0.692–0.926)

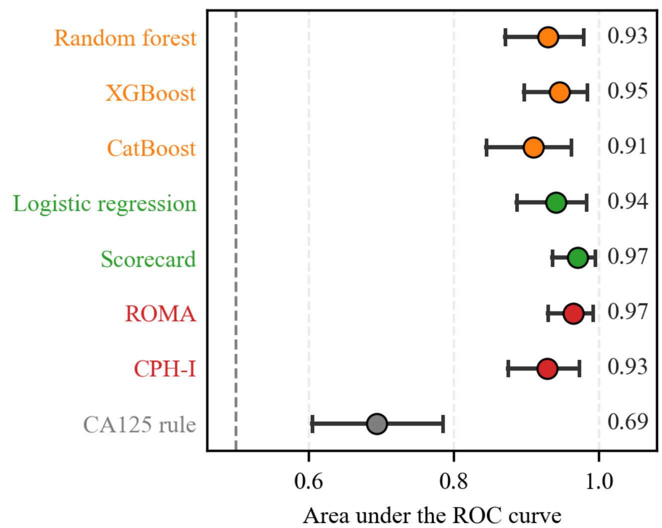

Fig. 1. External validation AUROC, 100 West China patients with measured HE4 (63 cancers); whiskers: bootstrap 95% CIs. Label colors: green = linear models, red = clinical formulas, orange = tuned trees, grey = simple rule.

B. Decision consequences

In the primary weighted-harm analysis (Q2; Table III, Fig. 2), every model was within-cohort recalibrated. The scorecard's minimum harm was below recalibrated ROMA and CPH-I at weights up to 5 (12.1 vs. 14.7 and 15.0 at $w = 1$); at $w = 50$, both recalibrated formulas fell below it (ROMA 45.8, CPH-I 50.3, vs. 59.5). The logistic regression was lowest up to $w = 10$; at $w = 50$, recalibrated ROMA was lowest (45.8). Reported minima are over the swept thresholds; where a minimum exceeds treat-all (e.g., XGBoost at $w = 50$), treating everyone dominates (Table III).

As the Q1 decision consequence (unchanged transport), the scorecard's high-risk tier (score $\geq +1$) missed 47 of 188 cancers with 5 false positives, versus 82 (ROMA) and 48 (CPH-I) at the

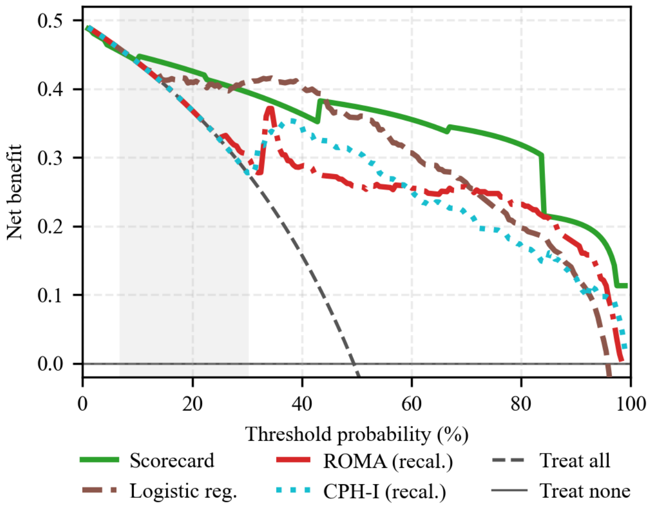

Fig. 2. Decision curve analysis on West China (N = 380): net benefit per patient (true positives minus threshold-weighted false positives) against the decision threshold in percent, with treat-all (dashed) and treat-none (the horizontal axis) references; the shaded band marks the published-cutpoint range (7–30%). The scorecard and the two formulas are shown after within-cohort recalibration. XGBoost is omitted; minimum harms are in Table III.

TABLE III

MINIMUM NET HARM ON WEST CHINA (N = 380) AT EXCHANGE-RATE WEIGHT w: HARM = w×MISSED+UNNECESSARY PER 100 WOMEN, MINIMIZED OVER THE SWEPT THRESHOLDS; TREATING EVERYONE COSTS 50.5 AT EVERY WEIGHT. LOWER IS BETTER. Q2 ROWS: SCORECARD, ROMA, AND CPH-I ARE WITHIN-COHORT RECALIBRATED; THE LOGISTIC REGRESSION AND XGBOOST ARE SHOWN AS FITTED. Q1 PUBLISHED-SCALE HARMS: ROMA 15.0/43.4/45.8/45.8; CPH-I 15.0/51.3/64.7/159.5.

Model	w=1	w=5	w=10	w=50
Logistic reg.	**9.7**	**29.2**	**39.7**	46.6
Scorecard	12.1	30.5	45.0	59.5
ROMA (recalibrated)	14.7	42.9	45.8	**45.8**
CPH-I (recalibrated)	15.0	50.3	50.3	50.3
XGBoost	14.7	44.7	66.1	215.5
Treat all (reference)	50.5	50.5	50.5	50.5

published cutpoints.

C. Calibration and missing-data checks

For calibration on the imputed cohort, the scorecard's logistic mapping was refit on West China; the logistic regression is shown frozen from training, and the formulas at published scales. Because the scorecard's fitting and evaluation share the cohort, these are apparent-calibration figures, not prospective ones, and only its figures can be inflated by the shared data. The recalibrated scorecard had Brier 0.097, calibration slope 1.00, and expected calibration error (ECE) 0.042; the logistic regression had Brier 0.111, slope 1.44, and ECE 0.135. The published formulas were farther from the data: ROMA had Brier 0.209, slope 1.53, and ECE 0.220; CPH-I had Brier 0.259, slope 1.11, and ECE 0.303. Both over-predicted risk (calibration-in-the-large −0.95 and −1.42).

Refitting without HE4 gave 0.925 (scorecard), 0.932 (logistic regression), and 0.891 (XGBoost). Multiple imputation by chained equations (MICE) preserved the flexible-model ranking (logistic regression 0.926) but degraded the scorecard's tertile binning (0.853).

D. Robustness and simulation

Under 30% multiplicative noise (West China, $N = 380$), losses were 0.040 for the scorecard, 0.002 for the logistic regression, 0.074 for ROMA, and 0.086 for XGBoost.

In the simulation (Figure 3), the ensemble-average minus-scorecard gap averaged +0.018 at drift level $d = 0$ of (1) and −0.018 at $d = 1$; at $d = 1$ the scorecard had the highest AUROC in 22 of 24 cells. Across the four simulation factors, the spread of the gap was largest across drift levels (0.033), followed by distribution shift (0.012), training size (0.003), and noise (0.002).

E. Internal validity and sensitivity

Across 20 repeated five-fold cross-validations, the scorecard's internal AUROC was 0.906 (SD 0.003), and the deployed three-bin configuration was selected in 18 of 20 repeats. In-sample the ensembles fitted at 0.991–1.000, falling 0.094–0.117 under CV.

IV. DISCUSSION AND CONCLUSION

The results show three things. First, on discrimination: greater model complexity did not consistently yield better external performance; seven integer parameters achieved external discrimination comparable to, or better than, substantially more complex models—the scorecard was numerically best on the complete cases (0.971), the logistic regression was highest on the imputed cohort (0.936 vs. 0.929), and CPH-I won the Japan stress test (0.920). Second, on decisions: the published cutpoints missed 82 of 188 cancers versus 47 for the scorecard's fixed high-risk tier, and with all models recalibrated its minimum harm stayed lowest up to 5:1. Third, on transportability: the simulation offers one possible explanation— flexible models can exploit interaction and nonlinear patterns that do not persist; when those patterns were removed, the scorecard had the highest AUROC in 22 of 24 cells.

The main limitations are as follows. (i) All three cohorts are Asian hospital series, so results may not transfer to other populations. (ii) HE4 was missing for 74% of West China patients and age for 85% of Japan patients, so much of the external comparison rests on training-set constants; and the Japanese cohort consists of healthy volunteers rather than women with masses, so its discrimination is not directly comparable with West China's; both narrow how broadly the findings apply. (iii) ultrasound-based models could not be tested (the public data lack imaging); (iv) the simulation is a simplification, not a proof about any specific hospital.

Future work should add ultrasound variables and validate prospectively in larger, multi-center cohorts; on these small cohorts the scorecard already performed no worse than far more complex models and stayed fully transparent.

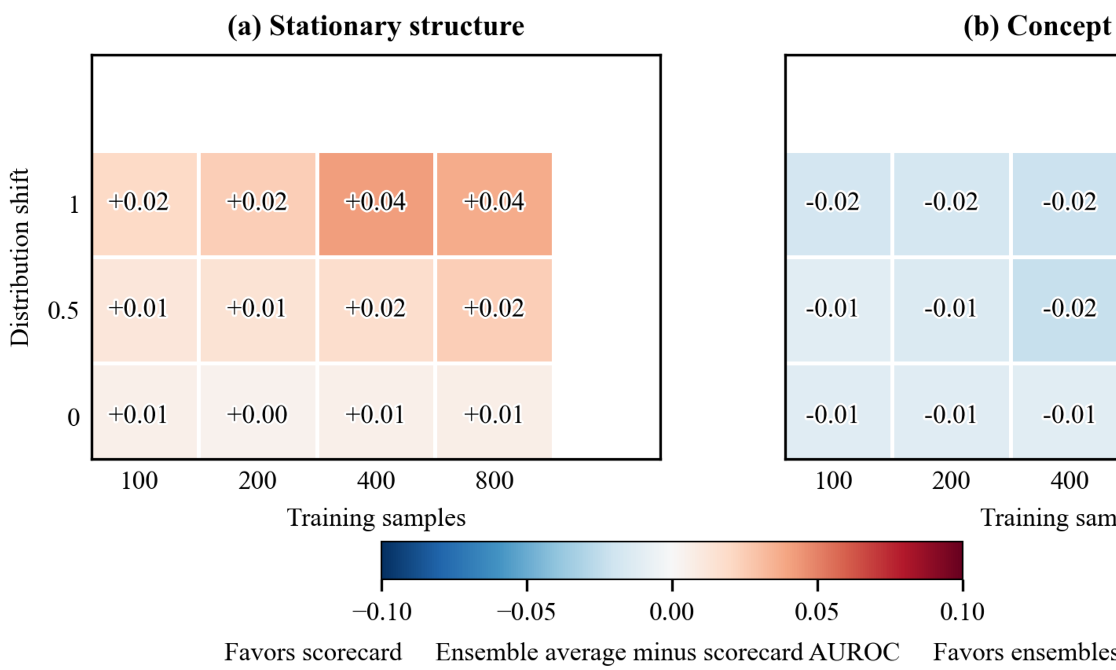

Fig. 3. Regime map from the simulation: each cell is the mean over 20 replicates of the ensemble-average minus scorecard test AUROC at 50% noise; red favors the ensembles, blue the scorecard. Axes: training samples $n \in \{100, 200, 400, 800\}$; distribution shift $s \in \{0, 0.5, 1.0\}$.

Acknowledgment

The authors acknowledge the use of Google Gemini, an AI-based assistant, for grammar and language checks and for basic code debugging; the authors retain full authority and responsibility for the design, methodology, analysis, and all claims of this paper.

References

[1] R. G. Moore, M. C. Miller, P. Disilvestro, et al., "Evaluation of the diagnostic accuracy of the risk of ovarian malignancy algorithm in women with a pelvic mass," *Obstet. Gynecol.*, vol. 118, no. 2, pp. 280–288, 2011.
[2] M. A. Karlsen, E. V. S. Høgdall, I. J. Christensen, et al., "A novel diagnostic index combining HE4, CA125 and age may improve triage of women with suspected ovarian cancer," *Gynecol. Oncol.*, vol. 138, no. 3, pp. 640–646, 2015.
[3] E. Kawakami, J. Tabata, N. Yanaihara, et al., "Application of artificial intelligence for preoperative diagnostic and prognostic prediction in epithelial ovarian cancer based on blood biomarkers," *Clin. Cancer Res.*, vol. 25, no. 10, pp. 3006–3015, 2019.
[4] E. Christodoulou, J. Ma, G. S. Collins, E. W. Steyerberg, J. Y. Verbakel, and B. Van Calster, "A systematic review shows no performance benefit of machine learning over logistic regression for clinical prediction models," *J. Clin. Epidemiol.*, vol. 110, pp. 12–22, 2019.
[5] L. Wynants, B. Van Calster, G. S. Collins, et al., "Prediction models for diagnosis and prognosis of COVID-19: systematic review and critical appraisal," *BMJ*, vol. 369, p. m1328, 2020.
[6] M. Lu, Z. Fan, B. Xu, et al., "Using machine learning to predict ovarian cancer," *Int. J. Med. Inform.*, vol. 141, p. 104195, 2020.
[7] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nat. Mach. Intell.*, vol. 1, no. 5, pp. 206–215, 2019.
[8] R. Caruana, Y. Lou, J. Gehrke, P. Koch, M. Sturm, and N. Elhadad, "Intelligible models for healthcare: predicting pneumonia risk and hospital 30-day readmission," in *Proc. 21st ACM SIGKDD*, 2015, pp. 1721–1730.
[9] Q. Mi, J. Jiang, T. Znati, et al., "Data for: Using machine learning to predict ovarian cancer," Mendeley Data, vol. 11, 2020, doi: 10.17632/th7fztbrv9.11.
[10] Li, "Comparative analysis of clinical characteristics between ovarian cancer and ovarian cyst patients," figshare, 2025, doi: 10.6084/m9.figshare.28831256.
[11] H. Machida, T. Hirakawa, K. Tsunekawa, T. Kimura, M. Murakami, and Y. Abe, "Revised cut-off value of human epididymis protein 4 enhances its use as an ovarian tumor marker," *Gynecol. Obstet. Invest.*, vol. 88, no. 6, pp. 349–358, 2023.
[12] A. L. D. Rolfsen, A. A. Dahl, A. H. Pripp, and A. Dørum, "Base rate of ovarian cancer on algorithms in patients with a pelvic mass," *Int. J. Gynecol. Cancer*, vol. 30, no. 11, pp. 1775–1779, 2020.
[13] T. Chen and C. Guestrin, "XGBoost: a scalable tree boosting system," in *Proc. 22nd ACM SIGKDD*, 2016, pp. 785–794.
[14] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: unbiased boosting with categorical features," in *Adv. Neural Inf. Process. Syst.*, vol. 31, 2018.
[15] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.
[16] B. Van Calster, D. J. McLernon, M. van Smeden, L. Wynants, and E. W. Steyerberg, "Calibration: the Achilles heel of predictive analytics," *BMC Med.*, vol. 17, p. 230, 2019.
[17] A. J. Vickers and E. B. Elkin, "Decision curve analysis: a novel method for evaluating prediction models," *Med. Decis. Making*, vol. 26, no. 6, pp. 565–574, 2006.
[18] M. Roberts, D. Driggs, M. Thorpe, et al., "Common pitfalls and recommendations for using machine learning to detect and prognosticate for COVID-19 using chest radiographs and CT scans," *Nat. Mach. Intell.*, vol. 3, no. 3, pp. 199–217, 2021.

Appendix A. Glossary of Terms

AUROC: area under the ROC curve; CA125: cancer antigen 125; HE4: human epididymis protein 4.